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**FINAL PROJECT REPORT**

**STATISTICAL ANALYSIS**

**TOPIC**

# Analyzing and Forecasting Household Electricity Consumption Using Statistical, Machine Learning, and Deep Learning Approaches

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## 1. INTRODUCTION

It is obvious to clarify that electricity plays an essential role in modern life and the socio-economic system. Additionally, understanding the factors affecting electricity consumption is important for load planning, demand forecasting, and grid operation optimization. Among which of them, including weather and especially temperature, has long been considered one of the most important forecasting factors (Pardo et al., 2002; Valor et al., 2001; Fan & Hyndman, 2012). However, the influence of temperature on electricity consumption is not always strong, especially in hourly load data. At this frequency, electricity consumption may be more influenced by load lag, daily life cycle, and system inertia (Hong, 2010; Hippert et al., 2001). Therefore, the impact of weather must be evaluated in parallel with the endogenous attributes of the time series. According to this research, which aims to evaluate the influence of weather, especially temperature, on hourly electricity consumption, and compare this influence with time variables (lag, rolling, seasonality). The data includes more than 10,000 hourly observations, with many meteorological variables and time series. The research method combines: Descriptive Statistics, Inferential Statistics, Regression Analysis, ARIMA, ARIMAX, SARIMAX, Random Forest, Decision Tree, LSTM, XGBoost Regression, Hybrid SARIMAX and LSTM. The main objective of this research support to determine the influence of temperature on hourly electricity consumption and compare it with the endogenous impact of the time string variables.

## 2. LITERATURE REVIEW

The relationship between temperature and electricity consumption has been recorded in many international studies. Pardo, Meneu & Valor found that temperature increases lead to the demand for electricity, mainly by using cooling systems. Similarly, Al-Hamadi & Solima have demonstrated that temperature is a crucial predictor in load prediction. Some research suggests the non-linear relationship between temperature and load. Fan & Hyndman indicated that load surged in approximately high temperatures, creating a non-linear structure, making linear regression ineffective. Others implemented LSTM, such as C. Cai et al. confirmed that deep learning can simulate this relation better. Regarding hourly data, Hong and Hippert et al. indicated that lag variables are more important than temperature because of the habit of consumption and the cycle of strong days. This makes temperature just play a supporting role in a short-term prediction.

# 3. RESEARCH METHODOLOGY

## 3.1 Descriptive Statistics

Descriptive statistics is a collection of methods to describe and summarize data to understand the basic characteristics of each variable before proceeding to the deep statistics. The indices such as mean, standard deviation, min-max, median, and percentiles help to show the trend of the center and the level distribution of data. Additionally, distribution shape measures such as skewness and kurtosis allow to recognize whether the data has deviation or outliers. This process plays an important role in assessing the reasonableness of data, detecting unusual data, and showing the right direction to proceed for the next model.

## 3.2 Inferential Statistics

Inferential Statistics is a group of methods used to conclude the population based on data samples. In the topic of household electricity data analysis, inferential statistics are used to test whether weather factors have a significant impact on electricity consumption. And compare power consumption between weekdays and weekends. Also, supporting to determination of whether the linear relationship (correlation) is statistically significant or not.

## 3.3 Linear Regression

Linear regression is used to analyze the effect level of the weather factor on the level of electricity consumption(active_power). Implementing this statistic to answer those main questions: Does weather factor make the electricity consumption increase?, Do humidity, pressure, and speed have a linear relationship with electricity consumption?. In this implementation, multiple linear regression was implemented to analyze the effects of all weather variables. This is an important part before implementing in other complicated models.

## 3.4 ARIMA Model

The ARIMA model is used as a simple time series model, focusing on assessing the capability of forecasting electricity consumption(active_power) based on the endogenous structure of the main series data, not using any weather variables. The target of ARIMA in this research includes checking the level of time dependency of the electrical load, identifying the baseline model to compare with advanced models, and evaluating whether the endogenous factors (lad, trend, noise) are strong enough to forecast the electrical load without considering weather information.

ARIMA includes three components:

AR(p): The level of current electricity consumption depends on the previous data.

I(d): number of steps to stop the series.

MA(q): modeling error noise of previous steps.

Model ARIMA is trained by applying all grid searches in all parameter sets and choosing the best model based on AIC, MAE, and RMSE. ARIMA is considered a baseline model in research, providing a comparable point to assess the improvement when adding weather variables(ARIMAX) or using machine learning or deep learning(Random Forest, LSTM).

### 3.5 ARIMAX Model

ARIMAX (ARIMA with Exogenous Variables) is a temporal sequence structure that expands from the ARIMA methodology, which allows additional variables of outer factors in the prediction process. It allows for the inclusion of additional external implements in the prediction process. This adds support to the model, as it does not just depend on past behavior but also considers outside factors influencing the target variable.

The ARIMAX model is applied after checking the stationarity of the series (ADF) and analyzing the autocorrelation structure (ACF/PACF). The steps of determining parameters of p, d, and q are performed using the ARIMA procedure, and then X part is extended. Finally, a series of ARIMAX configurations is tested by grid-search to select the optimal model, based on the forecast error (MAE, RMSE). The results show that ARIMAX is more suitable than ARIMA because it takes advantage of external information.

Therefore, in the forecasting problem of hourly electricity consumption, this model is suitable due to the electricity consumed dependence on many external factors, including weather ( temperature, humidity, pressure, etc), energy-consuming behavior in the previous hour and 24h cycle, and the environmental factors.

### 3.6 SARIMAX Model

The SARIMAX model is the more completing version compared to the ARIMAX model, because it is designed to additionally maintain the model seasonality. Moreover, with hourly electricity data, the ACF graph and FFT analysis show that the 24-hour cycle is very clear, so the electricity consumption repeats according to daily routine: low consumption in the morning, increased at noon, and highest consumption in the evening. Therefore, SARIMAX is used to add seasonal components ($P$, $D$,$Q$ ,$s$) where $s$ = 24 hours

Additionally, SARIMAX allows the model to learn the daily repeating pattern of which that ARIMA and ARIMAX cannot fully describe. When applied to electricity data, this model is expected to reduce the forecast error in periods with strong repeating structures, especially peak and off-peak hours.

In this study, SARIMAX is used after confirming the existence of seasonality using signal analysis methods. The model is then grid-searched with a reduced set of parameters to avoid excessive running time. The best model is then chosen based on the forecast error and the Log-likelihood goodness as AIC  of fit.

### 3.7 Random Forest Model

Random Forest is a machine learning model that includes many decision trees running simultaneously. Each tree is trained on a part of the data and a set of random variables, which brings less overfitting, better processing for multi-directional data, and captures the nonlinear relationship between variables.
For the electricity consumption by hour prediction topic, which contains strong fluctuations and nonlinear relationships, Random Forest is a more suitable option than the ARIMA/ARIMAX.

### 3.8 Decision Tree Model

A Decision Tree is a tree-like regression model that works by dividing the data into branches based on conditions that help reduce the variance of the response variable. Thanks to its intuitive branching structure, the model allows for easy interpretation of how the input factors affect the expected outcome.

In the hourly electricity load forecasting problem, a Decision Tree is suitable because it can learn nonlinear systems and take advantage of special levels such as ap_lag_1h, ap_lag_24h, and ap_roll_24h to describe the time dependence. This model also serves as a baseline nonlinearity for comparison with more complex methods in the study.

### 3.9. LSTM Model

Before talking about the LSTM model, we should briefly talk about the RNN model, a recurrent neural network that processes sequence data. However, RNNs have the "vanishing/exploding gradient" problem, which makes it difficult to remember long-term information such as hourly DATASETS like our group's. Therefore, Long Short Term Memory was born as a variant created to solve the problems of the Recurrent Neural Network (RNN), helping the model remember information for a long time.

The strengths of LSTM are its memorization mechanism, which forgets through gates such as input gate, forget gate, and output gate. Another ability is learning complex, nonlinear and cyclic patterns in data. Last but not least, it reduces gradient loss in long sequences.

### 3.10. XGBoost Regression Model

The XGBoost Regression model is used as a non-linear ensemble learning model (non-linear ensemble model). This model focuses on evaluating the predictability of the electric consumption (active_power) using both the time endogenous structure (lagged values, trends) of the time series and exogenous factors (weather and time).

The scope of Xgboost Regression in this study include:
1. Prediction: Predicting the actual electricity consumption (active_power) over an hourly timeframe (Hourly) which requires high accuracy.

2. Accuracy Evaluation: Assessing the accuracy of the prediction through statistical indicators (MAE, RMSE, MAPE, $R^2$).

3. Impact Identification: Identifying the importance (Feature_Importance) of exogenous factors (weather) and consumption history on active_power.

4. Advanced Model: Considered an advanced forecasting model (compared to linear models), providing a reference point for evaluating the level of performance improvement when using complex machine learning algorithms.

The XGBoost model is trained by temporally splitting the data into 80% of the data for training and 20% the rests for testing. The model hyperparameters are optimized by using the search technique such as Grid Search, and the model is evaluated on the testing set using MAE, RMSE, MAPE and $R^2$ to measure the predictive performance.

### 3.11. Hybrid SARIMAX-LSTM Model

The Hybrid SARIMAX-LSTM model is used as an advanced forecasting model (advanced forecasting model), focusing on taking advantage of the strengths of two methods: Linear Model (SARIMAX) and Nonlinear Model (LSTM). It works on the principle of time series decomposition, in which active_power ($Y_t$) is decayed into two components $L_t$ (Linear) and $N_t$(Nonlinear).

The goals of the Hybrid Model in this study includes:
Comprehensive Capture: Capture and Predict electricity consumption (active_power) by combining the strengths of two methods: linear model (SARIMAX) and nonlinear model (LSTM).

Improved Accuracy: Improve prediction accuracy compared to using only a single model (such as ARIMA or XGBoost), by decomposing and modeling different components of the time series.

Timeframe: Predict the average daily electricity consumption (Daily).

Validation: Verify the hypothesis of the existence and significance of both linear and nonlinear components in the electricity consumption time series.

The Hybrid model is trained in the splitting the data into 80% of the data for training and 20% the rests for testing and also in a two-step sequential process:

Train SARIMAX to generate residuals, then train LSTM on those residuals. The final prediction is the summary of the SARIMAX prediction and the LSTM prediction ($L_t + N_t$). The model is evaluated using metrics such as MAE, RMSE, and MAPE to quantify the eror between the hybrid prediction ($L_t + N_t$) and the actual active_power value on the test set.

## 4. IMPLEMENTATION AND EXPERIMENTS

### 4.1 Descriptive Statistics

#### 4.1.1 Formulas

**Mean**

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

**Standard Deviation**

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

**Coefficient of Variation (CV%):**

$$CV = \frac{s}{\bar{x}} \times 100$$

**Skewness**

$$\text{Skewness} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$$

**Kurtosis**

$$\text{Kurtosis} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4 - 3$$

### 4.1.2 Implement method

The process of descriptive statistics went through steps: first of all, the data was cleared by changing the time columns to the datetime type and eliminating missing values. Secondly, data was collected by hour to reduce the level of noise and create a more sustainable structure. Then, time variables such as weekday or is_weekend were created to support the analysis by cycle. The lag variables were added to catch the dependency by hour. In addition, rolling variables were calculated to show the average trend of the long term. Lastly, all variables were gathered through a descriptive statistics table with a distribution diagram, boxplot, and heatmap to provide an overview of the data.

## 4.2 Inferential Statistics

### 4.2.1 Formulas

The main methods include:

- t-Test (One-sample / Two-sample / Paired): to test the average difference in electricity between two groups.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

- ANOVA (Analysis of Variance): to test how much power consumption in multiple time groups such as in the morning, afternoon, and night.

$$F = \frac{SS_{between}/df_{between}}{SS_{within}/df_{within}}$$

- Pearson correlation test: support to measure the correlation between electricity and weather variables.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

- Chi-square test (if classified): to test the dependence between discrete variables.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

=> Generally, all tests above are based on p-value < 0.05 to conclude statistical significance.

**4.2.2 Implement method**

To prepare data, we create groups to compare between weekday and weekend, hours of the day (0–6, 6–12, 12–18, 18–24) and remove NA values. Then choose the test type including test objective, t-test to compare weekday vs weekend, ANOVA to compare consumption by 4 time frames, Pearson correlation to test weather effect using and Chi-square to test categorical variable. Next step, we set up a hypothesis by making the t-test, maintaining the null hypothesis ($H_0$) and the hypothesis that needs to be improved (H). Then, calculating statistics and p-value by using SciPy library, such as stats.ttest_ind, pearsonr, f_oneway, etc. Finally, it is necessary to draw conclusions according to the p-value

**4.3 Linear Regression**

**4.3.1 Formula**

$$Y_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + b_n X_{ni} + u_i$$

**4.3.2 Implement method**

The raw data was cleared and exchanged from date, month, and year type to datetime, then resampled the data to an hour to reduce data noise, which was recorded by minutes. The missing sample period was interpolated to ensure the continuity of the time series. From the main variable active_power, two characteristics of delay are created, including the capacity of 1 hour before and 24 hours before, helping the model to learn from the cycle of consumption. After eliminating missing variables, the data was divided into 3 parts: 85% to train, 10% to validate, and 5% to test. The model linear regression was trained on the training part and analyzed by the indices RMSE, MAE, and MAPE on the validating and test parts. Lastly, the model was trained again on all data train and validation before predicting on the test set to make sure optimal productivity.

### 4.4 ARIMA

#### 4.4.1 Formulas

$$Y_t = \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + .. + \beta_0 Y_0 + \epsilon_t$$

$$Y_{t-1} = \beta_1 Y_{t-2} + \beta_2 Y_{t-3} + .. + \beta_0 Y_0 + \epsilon_{t-1}$$

#### 4.4.2 Implement method

In the ARIMA model, data were processed and resampled to an hour to ensure the continuity of the time series. Then, the active_power was checked that it was stationary or not by the ADF test to identify the difference. When the series reached stationarity, ACF and PACF charts were used to suggest the scope value of p and q. Next, the process of calculating, which included 3 parts: train, validation, and test. Each ARIMA model grid search was performed on many structures of p, d, and q, and each model was evaluated on the RMSE of the validation set combined with AIC to choose the optimal model. After that, identifying the best parameters, the model was checked again on all data and validated, and performed a prediction on the test set to assess efficiency by MAE, RMSE, and MAPE. Lastly, the model was visualized by a graph to compare the real and predicted values.

### 4.5 ARIMAX

#### 4.5.1 Formulas

The ARIMAX(p,d,q) was illustrated with these formula:

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \beta^\top X_t + \varepsilon_t$$

By combining both external and internal information, the ARIMAX model is going to have better description of many complex relationships between power consumption and environmental factors

In the household electricity consumption dataset, it shows that electricity use varies significantly throughout the day. It is also influenced by external factors like temperature, weather, humidity, and wind speed. Moreover, there is a strong relationship with past values (lag 1 hour, lag 24 hours).

Therefore, ARIMAX is a good choice because it combines weather data and living patterns into the model. This helps to lower forecast errors compared to traditional ARIMA, which only uses the series' internal data. Also, since it is a linear model, it is easy to explain and fits well for the research paper.

### 4.5.2 Implement method

Firstly, we prepare data by converting time column to datetime, sorting by time. Then resample data by hour to ensure consistency. And create some exogenous features such as temperature, humidity, pressure, wind speed and lagged variables including lag 1h, lag 24h, and rolling mean. After that handle missing values if any, and standardize data format. The next step is to verify parameters p ,d and d by using ADF test to determine the required difference level to verify d. To verify q , we analyze ACF graph to estimate MA order. After that we also applying PACF to estimate AR order then indicate  parameter p.

Secondly, it is necessary to split data into train, validation and test parts. Due to our medium dataset, we prefer to split it into 85% train, 10% validation, and 5% test to optimize the accuracy.

The third step is training ARIMAX model by using the training set and exogenous variables to estimate model parameters. Then, apply grid-search strategy on validation set with p and q values in a predefined range. After running the grid-search strategy, we can choose the optimal model by comparing all ARIMAX models based on evaluation indices such as AIC, MAE, RMSE and select which model has the lowest error and highest fit.

The next step is to refit the optimal model on the entire 95% of the input data to increase stability. After that make a forecast for this model on the test set to check the generalization ability and calculate the MAE, RMSE, MAPE indexes. Finally, draw a comparison chart between the actual value and the forecast value.

## 4.6 SARIMAX

### 4.6.1 Formulas

SARIMAX (Seasonal ARIMAX) is an extension of ARIMAX by adding a seasonal structure:

$$\Phi_P(L^s)\phi_p(L)(1-L)^d(1-L^s)^D y_t = \Theta_Q(L^s)\theta_q(L)\varepsilon_t + \beta^\top X_t$$

Based on the dataset, it may be inferred that the daily electricity consumption in a 24-hour cycle increases in the evening, but decreases in the morning, needing to be modeled by seasonal AR + seasonal MA. Moreover, the weather and lag variables directly affect electricity consumption, must keep them as ARIMAX. Consequently, SARIMAX is more suitable than ARIMAX in a series with daily repetition.

### 4.6.2 Implement method

To prepare data for training model ,we resample data by hour, create lag variables 1h, 24h, and rolling 24h ,then type NA and set time as index. Also, normalize dataset to 1 hour cycle that suitable for 24h seasonality

Before running model ,we need to detect and determine seasonality by using those test . First is ACF test, used to determine strong spike appears at lag = 24. Sencond test is PACF, used to determine which has a cyclical structure. Lastly, we have FFT (Fast Fourier Transform) is used to generate a strong frequency peak at a period of ~24 hours. Therefore, from the above analysis, we can determine the seasonal cycle s = 24.

Then, the next step is to determine the non-seasonal components (p, d, q). Firstly, we already have d is taken from ARIMA/ARIMAX part (stationary series, d = 0) and p and q are estimated by PACF and ACF. However, due to the uncertainty, extending grid-search p=0..1 and q=0..1 to avoid overfitting and reduce running time

In additional, we have to determine seasonal parameters (P, D, Q) which have ever appeared in ARIMA or ARIMAX. Due to the dataset , the electricity series fluctuates by day, so choose seasonal differencing D = 1. And P, Q are tested from 0 to 1. Therefore, by combining a total of 16 models : (p,d,q) affiliates with {0,1} , (P, D, Q) affiliates with {0,1}, and s=24

About splitting the data on train, validation and test,we keep the same structure as ARIMAX with Train 85%, Validation 10% and Test 5%, to ensure the model evaluates the time correctly.

To train SARIMAX via Grid-search, it is iterated through all the merging groups between (p,d,q) and (P,D,Q,s). Then, modifying them to match this current model on train and calculating RMSE, AIC to rank the models. In order to choose the best model, we prioritize the model with the lowest RMSE (best prediction) and AIC is used to select the less complex model.

After that we retrain the optimal model on train and validation by using 95% of the initial data, aim to increase stability before evaluation.Final is the main step that make a evaluation on Test set including prediction rely on refit model, calculating MAE, RMSE, MAPE and visualizing them by drawing the Actual vs Forecast chart for comparison.

**4.7 Random Forest**

**4.7.1 Formulas**

Random Forest is an ensemble model that combines multiple Decision Trees, where each tree is trained on a bootstrap sample of the original data. The final prediction is calculated by averaging the predictions of all the trees:

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

In each tree, the split point is chosen based on the variance reduction:

$$Gain = Var(parent) - Var(split)$$

With variance given by:

$$Var(t) = \frac{1}{N_t} \sum_{i \in t} (y_i - \bar{y}_t)^2$$

### 4.7.2 Implement method

The Random Forest model is trained in a process consisting of main steps: first, the data is cleaned, re-indexed and resampled hourly to ensure the continuity of the time series; then important features such as 1-hour lag, 24-hour lag, 24-hour moving average and time variables (hour, weekday, weekend) are created to describe the periodicity and time dependence of the electricity load. The processed dataset is divided in time order into Train (85%), Validation (10%), and Test (5%), then normalized using StandardScaler to ensure uniformity between models. In the tuning step, 40% of the training set is used for GridSearchCV combined with TimeSeriesSplit to find the optimal set of hyperparameters (number of trees, depth, number of leaf samples), helping the model achieve the lowest MAE while still ensuring the calculation speed. Finally, the optimized model is retrained on the entire Train + Validation set to maximize the data exploitation and evaluated on the Test set using MAE, MAPE, RMSE, and $R^2$ metrics to check the generalization ability of the Random Forest model.

## 4.8 Decision Tree

### 4.8.1 Formulas

A Regression Decision Tree is based on finding the split point that reduces the variance of the target variable. Each node is split according to the criteria:

$$Gain = Var(parent) - Var(split)$$

With the variance in the node given by:

$$Var(t) = \frac{1}{N_t} \sum_{i \in t} (y_i - \bar{y}_t)^2.$$

### 4.8.2 Implement method

The Decision Tree model was trained on the processed and hourly resampled dataset. After cleaning the data and interpolating missing values, important features such as 1-hour lag, 24-hour lag, and 24-hour rolling features were created to capture the time-dependent nature of the electricity load, in addition to temporal features such as hour, day, and weekend. The data was then reduced to only the columns that were actually needed, and rows containing NA values due to time shifts were removed. The dataset was split chronologically into Train (85%), Validation (10%), and Test (5%), and then normalized using StandardScaler to standardize the data scale. The Decision Tree model was fine-tuned using GridSearchCV combined with TimeSeriesSplit, with hyperparameters such as max_depth and min_samples_leaf, to find the configuration with the lowest MAE. The optimized model is then used to predict on the test set, evaluated using MAE, RMSE, and $R^2$, and the feature importance is analyzed, and the prediction results are visualized against the actual values.

**4.9 LSTM**

**4.9.1 Formulas**

An LSTM unit consists of four main equations:

1.Forget Gate

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

2.Input Gate

a, Input gate activation

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

b, Candidate cell state

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C)$$

3.Cell State Update

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

4.Output Gate

a, Output gate activation

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

b, New hidden state

$$h_t = o_t \odot \tanh(C_t)$$

### 4.9.2 Implement method

Data is divided in the correct time order (no shuffle):

• 85% train

• 10% validate

• 5% test

LSTM architecture used in the study:

•First layer we will run LSTM with 64 units

•Then dropout 20%

•The second layer will run LSTM with 32 units

•Then dropout 20%

• Dense for one step prediction output

Model is trained in:

• 20 epochs

• Batch size = 32

• Optimization using Adam

• Loss function: MSE/MAE

## 4.10 XGBoost regression

### 4.10.1 Formulas

$$\hat{Y}_t = \sum_{k=1}^{K} f_k(\mathbf{X}_t)$$

### 4.10.2 Implement method

The training and evaluation process of the XGBoost Regression model for the electricity consumption prediction problem (active_power) is is performed through a strict process, starting with the processing and transformation of the raw data. Specifically, the data can be read and resampled (Hourly Resampling) to reconstruct a continuous time series with the desired expected time frame.Next, instead of checking the stationarity as in the classical models, the process is placed into the Feature Engineering, where explicit independent variables are created to provide information about periodicity (weekday variation weekday, is_weekend), autocorrelation features (such as ap_lag_1h and ap_lag_24h), and historical point (ap_roll_24h), also combined with exogenous variables (weather) to form a set of input factors (X). Then, the data is analyzed in time order into training (80%) and Test files (20%) to ensure the reliability of the forecasting process of the model. The XGBoost Regression model is initialized and trained on the training set by using some optimal parameters (e.g., n_estimators, max_depth) to minimize the loss function. Finally, the model is evaluated on the test set to access the forcasting performance by using standard metrics such as MAE,RMSE, $R^2$ and MAPE, while the Feature Importance is analyzed to identify which predictors have the strongest influence.

### 4.11.Hybrid SARIMAX-LSTM

#### 4.11.1 Formulas

$$\hat{Y}_t = \hat{L}_t + \hat{N}_t$$

#### 4.11.1 Implement method

The hybrid model training process is performed in 2 sequential stages to take advantage of both SARIMAX ans LSTM model.

In the first stage, we begins with data processing and daily resampling (Daily Resampling) to ensure the continuty of the time series, then the data is split over time into Training set and Testing set. The SARIMAX model (with order (1, 1, 1) times (1, 1, 1, 7)) is trained on the Training set with exogenous variables (temp, humidity) to predict the linear component ($L_t$). Next, the residual ($N_t$) is computed by subtracting the actual active_power value from the SARIMAX prediction $L_t$ on the Training set.

In the second stage, focusing on training the LSTM model to predict the non-linear errors $N_t$. The residuals and exogenous variables are normalized ($MinMaxScaler$) and sorted into a time-windowed input structure (look_back=7) to train the Stacked LSTM model to minimize the mean squared error (MSE).

Finally, hybrid prediction on the Test set is performed by predicting $L_t$ and $N_t$ for each time step, then summing the results ($Y_{t\_predicted} = L_{t\_predicted} + N_{t\_predicted}$). The final performance of the model is evaluated using the MAE, RMSE, $R^2$ and MAPE metrics, and visualized to compare with the actual values.

## 5. RESULTS AND DISCUSSION

### 5.1 Descriptive statistics

The result showed that active_power had high fluctuations (CV high) and right-skewed distribution, meaning that the majority of consumption time is at the normal level, but sometimes there appear hours that have a high level of consumption. Conversely, variables such as temp and feels_like were near normal distribution, sustainable fluctuation, and fewer outliers. Variable speed was slightly to the right so that the wind was almost slightly. Additionally, time variables such as weekday and is_weekend were distributed by week. Lag and rolling variables following the active_power value, reflecting the strong cycle in electricity consumption. Overall, the data was clear, had a good structure, and just had some unusual consumption variables, but was absolutely suitable with real data and useful for the next predicted model.

### 5.2 Inferential statistics

#### 5.2.1 Pearson Correlation Analysis

The result shows that temperature (temp, feels_like) has the strongest positive correlation (~0.43). Indicate that when in the hotter the weather, households use more electricity, mainly due to cooling devices (fans, air conditioners). Furthermore, the humidity and pressure have a slight negative correlation. This means when humidity is high/pressure is high, energy consumption tends to decrease slightly. However, the degree is weak, so it is not the main factor. And the wind speed has a very slight positive correlation (0.16), so the effect is low, almost insignificant. Additionally, the p-value = 0.000 (< 0.05) for all variables means all the correlations are statistically significant.

To sum up, the weather factors have an impact on electricity consumption, of which temperature is the strongest factor. This is consistent with the actual behavior of using cooling devices.

#### 5.2.2 Scatterplots & Regression Lines

The picture illustrates that temp and feels_like have a clear linear trend with active_power, while humidity and pressure have a slightly negative slope regression line shows a small and decreasing effect. Also, the speed has a large dispersion, very slightly positive regression line. Therefore, the linear dependence of electricity on weather is real but not strong except for temperature.

#### 5.2.3 T-test between Weekday vs Weekend

As a result of this algorithm, p-value = 0.2014 (> 0.05), which means no statistically valid difference between weekday and weekend electricity consumption.

The barplot also shows that the consumption is almost the same, about 280–290W on average. Therefore, households have a relatively stable living standard between weekdays and weekends, not much affected by work/leisure schedules.

#### 5.2.4 ANOVA test for comparison of 7 days of the week

The boxplot shows that both Sunday and Monday-Tuesday have higher average consumption than other days. The period from Wednesday to Thursday is usually lower. And during the day, the fluctuations are very large due to many outliers, but the medians are quite different. Therefore, although the weekend is not different from the weekday, each day of the week is still different, reflecting the scheduled lifestyle habits (e.g., late sleep on weekends, using many devices at night leads to increased peak consumption.

### 5.2.5 Conclusion

Consequently, the temperature is the strongest weather factor that leads to increasing electricity consumption. While humidity, pressure, and wind have a very weak influence, they do not play a major role. However, there is no difference between weekday and weekend (t-test), but a difference appears between weekdays (ANOVA), proving cyclical living behavior.

## 5.3 Linear Regression

In this research, a linear regression model was actually a multiple linear regression because it used many input variables, including temperature, humidity, speed of wind, ap_lag_1h, and ap_lag_24h. Those variables play an important role in anticipating active_power in the present time. The result was assessed on the validation set, which showed RMSE of about 75.46, MAE of about 58.26, and MAPE of about 30.94%. This reflected that the model ran relatively well in waiting for the confirmation period, thanks to the ability to utilize the information from weather temperatures and late values of capacity series. When the model was applied to the test part, the error increased slightly, with RMSE at 77.88, MAE at about 60.55, and MAPE grew to 39.02%. This showed that the predictive model still remained at a sustainable level, but the accuracy decreased because of test set was highly fluctuating and contained more volatility than the validation set.

Predictive chart showed that line prediction based on the general trend of real data, particularly on the period of change, followed by an hour. However, the model could not catch up with the high fluctuation and peak or sharp bottom, which are popular characteristics of electricity consumption data. This is the natural limitation of the linear regression model, which is applied to time series with much noise and high fluctuation. Conclusion, multiple linear regression showed quite well results on the data by hour, and the prediction was reasonable at the trend level and light fluctuations, but limited in catching the sudden fluctuation of active_power

## 5.4 ARIMA

The result Augmented Dickey-Fuller (ADF) for active_power series showed that ADF Statistic by -6.2986 and p-value approximately 3.45e-08, less than 0.05. This allowed us to reject the assumption that this series was not stationary and conclude that the initial series was stationary. At the same time, the ADF Statistic value was less than all critical values at a significant level of 1%, 5% and 10%, continuously confirming the stationarity of the series. Because the series was stationary at first, the ARIMA model did not need to take differences. So, the optimal difference $d = 0$. Based on the PACF chart, we see a clearly outstanding point at lag 1, while other lags then decreased rapidly to 0 and did not exceed over significance threshold. This showed that autoregressive components (AR) primarily appeared at level 1. Because PACF "cut-off" strongly at lag 1, the AR level was suitable for the model with $p = 1$. With chart ACF, values decreased following the tailing type, but it did not have a clear-cut-off at any specific lag. ACF did not stop suddenly and remained relatively positive at many lags, which showed that the structure of the MA was not strong or clear. Because ACF did not see any clear cut-off, the MA level was suggested to be small, usually starting to test with q = 0 or q = 1. From the results of the 2 charts, PACF cut off at lag 1, so that we may choose p = 1, and ACF did not have a clear cut-off, so that we should choose q = 0 or 1. The first suitable ARIMA(1,0,q), where q was identified by testing and assessing the error of the model. Train/Validate/Test(Time Series Split) showed the process of splitting data following the characteristics of time series, whereas the data point was split based on time order. About 85% initial data was used for training, 10% the next data for validating the model in the process of choosing parameters, and 5% of the last data for testing to evaluate the ability of predicting real values. This division ensured the model always learned from the past and anticipated for future. The result of the Grid Search illustrated that the model ARIMA(2,0,1) had the lowest RMSE and simultaneously reached the lowest AIC, compared to other models. So it was considered to be the most optimal model. The best model ARIMA(2,0,1) was different from the initial prediction ARIMA(1,0,q) often happened because analyzing ACF/PACF suggested visualization. Charts have not always shown all of the structures of series, particularly data noise or having added elements. Grid Search and Auto ARIMA evaluated the direct productivity of the model based on real data, so that it could detect 1 another AR( p = 2) helps the model to anticipate more correctly. So, ARIMA(2, 0, 1) was chosen due to the lowest error on the validation set.

After training the model ARIMA(2,0,1) on all set training and validation, the model was used to predict on the test set. The result showed that MAE on test was about 95.84, RMSE was 110.96, and MAPE was about 73.37%. Those errors reflected that the model could have the ability to catch the trend of the overall, but the absolute precision was still limited, particularly due to the high fluctuations and high noise in the active_power series. Model predictions showed that line anticipation remained a slightly increasing trend and more smoothly than the real data, while the data test fluctuated strongly over time. This means the ARIMA model had the trend to smooth the series and was not able to respond rapidly to sudden fluctuations of consumption capacity. However, the model still described the overall level and scope of variation of data. Conclusion, ARIMA(2,0,1) worked as expected with a series having a high level of noise, but it was still limited to anticipate the rapid variation, which was suitable with characteristics of a linear model such as ARIMA.

## 5.5 ARIMAX

### 5.5.1 Grid search ARIMAX

Based on the ARIMAX model training process described above , the team performed a grid-search with 16 different ARIMAX models, combining parameters p = 0..3 and q = 0..3. The grid-search results show that Models with high AR or MA structure ($p \geq 2$ or $q \geq 2$) often have low AIC but high validation RMSE → the model is too complex, easy to overfit. The simple models such as ARIMAX(0,0,0) and ARIMAX(2,0,2) give close RMSE, however, ARIMAX(0,0,0) gives more stable results and less fluctuations over many runs. The optimal model is selected according to the RMSE → AIC criterion: ARIMAX(0, 0, 0). Because the lowest RMSE validation among all simple ARIMAX models is combined with the small MAE validation. Also, the acceptable AIC and the simple model indicate that the model is stable, not overfit.

### 5.5.2 Model Interpretation

After selecting the best ARIMAX model, it is observed that ARIMAX(0,0,0) model was refitted on the entire train and validation set, which goes to 95% of the data. The summary results illustrate that the exogenous variables with strong significance, such as temp, ap_lag_1h, and ap_roll_24h, have very large coefficients and p-value < 0.01. This means it direct and significant impact on electricity consumption. Moreover, some of the weather variables have small coefficients and p-value > 0.05, showing the weak and consistent impact on reality, so electricity depends more on living activities than weather. Also, the noise (sigma$^2$) is large, and this reflects the characteristics of the electricity dataset: consumption fluctuates strongly by the hour, difficult to predict completely linearly.

### 5.5.3 Forecasting performance on test set

-MAE = 56.78 show model error is on average ~57 Wh per hour → acceptable error level.

-RMSE = 75.03 shows power fluctuates strongly → large errors in peak hours are normal.

-MAPE = 33.52% shows a percentage error greater than expected, proving that ARIMAX still does not describe complex fluctuations well (especially in the evening or on weekends).

The figure shows that the ARIMAX sticks quite well in low-medium consumption hours.Additionally, the forecast is biased when there are spikes (air conditioning, cooking, using high-power appliances). Therefore, this is true to the nature of ARIMAX: the linear + exogenous model cannot describe sudden on/off of appliances.

### 5.5.4 Discussion

a) Strength

- Simple, easy-to-interpret model, so it is easy to understand how weather + lag impacts power.

- Fast training, easy tuning

- Predicts are pretty good at normal consumption levels

b) Limitations

- Unable to capture power consumption spikes due to ARIMAX linear.

- Strong seasonality (24h) is not described because the clear day-night cycle is not well modeled.

-High MAPE leads to the percentage error not being optimal. This is the reason why the team had to switch to SARIMAX testing in section 5.7 to model more seasonal structure.

### 5.6.5 Conclusion for ARIMAX

Consequently, we can conclude that ARIMAX has improved over ARIMA by adding weather and lag factors. However, the model does not describe the daily repeatability of electricity consumption. The RMSE is still high, so we found that we need a model with a seasonal structure. Therefore, SARIMAX is chosen as a reasonable extension step to evaluate whether the error can be reduced or not.

### 5.6 SARIMAX

After completing the parameter determination step using ADF, ACF, PACF and seasonality testing, SARIMAX is deployed with a reasonable set of parameters and the optimal model is searched through a reduced grid-search.

#### 5.6.1 Grid-search SARIMAX

Based on the results of the seasonal analysis, the model uses a seasonal period S = 24 to evaluate on the Validation set. The forecasting targets prioritize the lowest RMSE, since RMSE reflects the forecast error in units of Wh/h, so the model was chosen to refit is:

$$\mathbf{SARIMAX}(0,1,1) \times (0,1,1)_{24}$$

=>This model is suitable due to its simple structure, strong Seasonal MA(1), which is suitable for 24h data, and has a stable performance

#### 5.6.2 Refit the best SARIMAX model

This result show that the electricity consumption is used in a daily repeating pattern, so it suitable for SARIMAX. Moreover, some weather variables have a certain impact but not too much. Specifically, the 24-hour cycle plays the most important role in this model.

#### 5.6.3 Forecast on test set

After making a forecast on the test set , the result shows that RMSE = 71.23 is the lowest in the whole model (ARIMA ~110, ARIMAX ~75). And MAE ~54.37 Wh/h is very low error per unit time. Also, MAPE is 30.23% meaning that the model's forecasts are off by an average of ~30% compared to the actual value, which is an acceptable margin of error for electricity data that fluctuates hourly.

From the results, we conclude that SARIMAX outperforms ARIMAX and ARIMA by capturing 24-hour seasonality. The electricity is forecasted well at most times, especially during peak and evening hours. And when there is a change in temperature, the model maintains an accurate trend due to the seasonal component and electricity lag.

#### 5.6.4 Discusion about SARIMAX model

a)Strength

-SARIMAX provides better forecasting performance than ARIMA and ARIMAX by modeling the day-night cycle (S = 24) of electricity consumption.

-The seasonal MA(1) component has a coefficient close to -1, indicating that the 24-hour pattern repeats very strongly and the model has learned this.

-SARIMAX also retains the advantage of ARIMAX when incorporating exogenous variables (temperature, humidity, pressure, etc.), helping to adjust the forecast according to weather conditions. Thanks to that, the RMSE is reduced to ~71, the lowest in the ARIMA-based group.

b) Limitation

-SARIMAX requires much longer training time than ARIMAX because of the additional seasonal parameter, and is prone to convergence warnings when optimizing.

-Due to this is still a linear model, so it is difficult to detect complex nonlinear behaviors in electricity consumption.

**5.6.5 Conclusion of SARIMAX model**

In summary, it can be stated that SARIMAX is the most effective model in the ARIMA based group, thanks to the direct modeling of the 24-hour cycle, which is the most prominent feature of the hourly electricity series.

The forecasting performance is:

+ RMSE ≈ 71.23

+ MAE ≈ 54.37

Therefore, this proves that the seasonal variable plays an important role in capturing the daily recurring electricity consumption behavior, helping SARIMAX outperform ARIMA (no exogenous, no seasonality) and ARIMAX (exogenous but no cycle modeling). Consequently, we can conclude that SARIMAX is considered the most optimal model in the time series forecasting part of this study.

**5.7 Random Forest**

The Random Forest model was fine-tuned using GridSearchCV combined with TimeSeriesSplit on the hourly resample dataset. The tuning used 40% of the traning data which help to reduce the processing time but the results still showed that the best performing model configuration was n_estimators = 150, max_depth = None, and min_samples_leaf = 2, with an MAE of about 5.58 on the tuning evaluation set. The differences between the hyperparameter sets were not too large, but the optimized model tended to achieve better results when allowing the tree to grow deep (max_depth=None) but was still tuned with min_samples_leaf to avoid overfitting. On the validation set, the model achieved MAE = 2.15, MAPE = 1.08%, RMSE = 3.14, and $R^2$ = 0.9988. These metrics show that the model simulates the hourly electricity load variation well, with a low average error and a very high coefficient of determination. The small error and high fit show that Random Forest makes good use of the lag and rolling features as well as the weather and time variables. When retraining the model with the entire Train + Validation set and evaluating on the Test set, the obtained metrics remain stable with MAE = 3.03, MAPE = 2.25%, RMSE = 4.65 and $R^2 = 0.9989$. The slight increase in error compared to validation is reasonable because the Test set has never been used in training. However, this increase is small and does not affect the overall reliability of the model. This confirms that Random Forest has good generalization ability and is not overfitting during the learning process. The Feature Importance results show that variables directly related to power such as apparent_power, current, power_factor and reactive_power play the most important role in forecasting. In contrast, time variables and lag features have a lower importance. This is consistent with the nature of load data, where the current power value is strongly influenced by the instantaneous electrical components. The Actual vs Predicted graph illustrates that the model closely follows the real data in the first 300 samples of the Test set. The load peaks, fast fluctuation regions and deep dips are all reproduced by the model with small delays and errors. This shows that Random Forest not only achieves good performance in terms of data but also accurately simulates the signal shape of the time series. In summary, Random Forest shows outstanding performance in the hourly electricity load forecasting problem, with strong nonlinear learning ability, high accuracy and good stability. This is the model with the highest performance among the models surveyed in the study.

**5.8 Decision Tree**

The Decision Tree model was trained on the hourly resample dataset. The dataset was split in time order with 85% for training, 10% for validation and 5% for testing, corresponding to 8678, 1021 and 511 observations, respectively. The hyperparameter search was performed using GridSearchCV combined with TimeSeriesSplit (5 splits). The experimental values were max_depth $\in \{5, 8, \text{None}\}$ and min_samples_leaf $\in \{1, 2, 4\}$. The optimal parameter set was obtained with max_depth = 8 and min_samples_leaf = 4. The evaluation results on the test set gave the MAE = 11.95, RMSE = 15.60 and $R^2 = 0.9777$. The model achieved a relatively high fit, but the error was still large at times of strong load fluctuations. The results of the analysis of the importance of the features showed that the variable apparent_power had the highest proportion. The remaining features such as hour, current, ap_roll_24h and ap_lag_1h had a lower contribution. The Actual vs Predicted chart shows that the model closely followed the general trend of the data in the first 300 samples. The error increased at load peaks, consistent with the limitations of the single decision tree model.Overall, the Decision Tree achieved average results, suitable for use as a reference model. The performance was lower than that of ensemble models under the same training conditions.

### 5.9. LSTM

In this LSTM model is trained using previous 24 hours of operation to predict power consumption in next 1 hour.

Evaluation results:

- MAE = 55.8 → The average error in the prediction times is about 55, which is not too large and is acceptable.
- RMSE = 73.3075 → RMSE higher than MAE indicates that the model has difficulty predicting periods of high electricity usage.
- MSE = 5373.9849 → suitable for highly volatile data

This is consistent with the characteristics of LSTM: Learning trends and cycles and difficult to accurately capture unusual, sharp peaks.

### 5.10.XGBoost regression

In this study, the XGBoost Regression model is applied to forecast hourly electricity consumption (active_power). As a non-linear and ensemble learning model, XGBoost regression is expected to outstandingly traditional linear models by capturing complex and highly non-linear relationships in noisy time series data. The model is trained on weather variables (temp, humidity, speed), time variables (weekday, is_weekend), and important lagged variables (ap_lag_1h, ap_lag_24h, ap_roll_24h). Performance analysis on the Test Set shows that the model achieved an MAE of 56.66, an RMSE of 74.94, and an MAPE of 31.064%. The model also shows that the $R^2$score reached 0.3987, indicating that the model explains roughly 40% of the variation in $active\_power$—a significant improvement over a simple mean-based baseline, especially given the extremely volatile and noisy nature of the dataset. At the same time, the Feature Importance analysis confirmed the model's validity, as the prediction was most strongly influenced by short-term inertia ($ap\_lag\_1h$: 48.34%) and daily cyclic behavior ($ap\_roll\_24h$: 35.55%), demonstrating that XGBoost was successful in effectively utilizing this historical information (accounting for nearly 84% of the weight). In summary, the XGBoost Regression model demonstrated its ability to fit highly noisy data, with reasonable predictions at the trend level and the ability to capture non-linear relationships, providing a solid foundation for further electricity forecasting studies.

### 5.11. Hybrid SARIMAX-LSTM

The Results and Discussion of the Hybrid SARIMAX–LSTMX model are built based on the daily averaged active_power energy consumption time series, by using temp and humidity used as exogenous variables. The linear component is modeled by SARIMAX with Order (1, 1, 1) and Seasonal Order (1, 1, 1, 7) which reflecting weekly seasonality. This step separates trend and seasonality and computes the residuals ($N_t$), which represent the nonlinear component.Next, the nonlinear component is then modeled using the LSTMX —a stacked LSTM network using a sliding window of $look_back = 7$days, trained on the residual series $N_t$ by using a sliding window of $look_back = 7$days and combined with the normalized exogenous variables. Finally, the hybrid forcast is obtained by summing the linear predictions ($L_t$) and the nonlinear estimates ($\widehat{N}_t$).However, the evaluation results on the test set show that the model has extremely poor performance, with the following error metrics: MAE = 84.693, RMSE = 96.313, MAPE = 48.314%, and especially $R^2$score is –2.913. A strongly negative $R^2$indicates that the model is performing more worse than simply predicting the test-set mean, meaning it fails to capture the structure of the smoothed time series.

## 6. OVERALL FINDINGS AND CONCLUSION.

### 6.1 Conclusion

After implementing the research to analyze and predict household power consumption in northeastern Mexico using a 14-month dataset with one-minute recordings, integrating with many weather factors such as temperature, humidity, pressure, and wind speed. The dataset also captures both the technical load characteristics and the meteorological conditions impact of energy usage. We applied both statistical models and machine learning models involving Linear Regression, ARIMA, ARIMAX, SARIMAX, Decision Trees, Random Forest, XGBoost, as well as many deep learning models like LSTM and a Hybrid model. As results are observed in Table IV that Random Forest is the best predicting model with MAE = 3.03, RMSE = 4.65, MAPE = 2.25%, and $R^2 = 0.9980$, which are the highest among all models. This indicates that Random Forest effectively models nonlinear relationships and interactions between energy consumption and weather factors. It can be said that both ARIMAX and SARIMAX models also performed well, especially in capturing trends and seasonality, but their accuracy was still lower than Random Forest. While deep learning models like LSTM can not exceed traditional machine learning models, because the limited timeframe in 14 months and the fact that electricity load depends more on external variables than on long-term time series variables. Generally, the results suggest that Random Forest is the most suitable model for forecasting household power usage in this dataset's context, because it give the low errors, stability, and simplicity in implementation.

### 6.2 Insights from the Study

From the process of analyzing data and comparing with predictive models, some important findings were identified. First of all, the level of electricity consumption had a significant relation with weather factors, including temperature, humidity, pressure, and speed of wind. When weather variables were included in the model, the accuracy of predictions obviously increased, particularly the error of MAPE decreasing from 73% in ARIMA to 33% in ARIMA and 30% in SARIMAX. This showed that the condition of temperature was a significant factor in the behavior of electricity consumption, particularly on hot days when the demand for electricity increased. Secondly, the data by hour showed the cycle of electricity consumption, with high electricity consumption in noon and in the evening, and the difference between weekdays and weekends. Those characteristics reflected the habit of using electricity in a household and showed that the time variables in the day were a crucial factor in prediction. Lastly, resampling data from minutes to hours helped stand out the key relation between weather, habit, and the level of consumption, improving the efficiency of the model.

### 6.3 Limitations

Although the study achieved many positive results and provided valuable findings, there are still some limitations that need to be noted. Firstly, the biggest limitation comes from the dataset itself. Furthermore, the data was recorded from a single household in the Northeast of Mexico, so the electricity consumption reflects the characteristics and daily behavior of this household, and it is not able to be generalized to other groups of households with different sizes, structures, income levels, types of appliances used, or socio-climatic conditions. This reduces the generalizability of the model. In addition, the dataset does not include important socioeconomic variables such as the number of members in the household, the level of use of electrical appliances, income, or information about holidays, vacations, or special events that may affect electricity consumption behavior. The lack of these variables makes it impossible for the model to assess the impact of macro and micro factors on electricity consumption. Another limitation of the data is to achieve the weather information from the OpenWeather API, with the purpose to measuring errors compared to the actual climate conditions at the household location. Therefore, it can affect the quality of the forecast, especially for models that are sensitive to weather variables. Moreover, the conversion of data from minute to hourly frequency, although necessary to reduce noise and stabilize the time series, has lost some short-term features, especially sudden fluctuations that occur in very short periods that the model can no longer capture. In addition, the analysis and modeling process also have certain limitations. Moreover, the research mainly focuses on a group of statistical and machine learning models; Other advanced techniques such as transformer models for time series and attention mechanisms have not been fully exploited. At the same time, the feature selection process has only stopped at the level of experience and basic statistical analysis, without implementing system optimization methods or sensitivity analysis to test the influence of each variable. Also, the model has not assessed the impact of seasonal changes in data structure or socio-climate shocks, which are important factors in energy economic analysis. Finally, the study has not conducted a detailed comparison of the stability of the model over different time periods or a strong test of stationarity and time series structure for all variables. Consequently, this limitations show that the research results need to be viewed within the scope of the data and methods used, also open up many directions for improvement for future research.

### 6.4 Recommendations

Based on the findings above, some practical suggestions could be recommended to optimize the management and prediction of electricity. With a regulatory agency and policy planner, identifying the significant effects of weather and time on the level of electricity consumption, allowing for creating policy to coordinate the electricity supply by season, particularly supplying more on hot days or periods that require high demand for electricity. Additionally, the result of the research also suggested the capability of applying the electrical price by hour(Time-of-Use Pricing), where the price will be higher in the peak hour or lower in the midnight or the hour when the electricity demand is low. This helps to regulate the behavior of using electricity and decrease the pressure on the electrical system. With the electrical company, a Random Forest model with outstanding efficiency could be implemented in a short-term prediction system by hour to enhance the regulatory capabilities and balance the load. Moreover, the integration of predictive models in the electricity smart grid could help to find the abnormal detection, optimize the distribution, and enhance the sustainability of the electrical system. In the household scope, users could utilize the predictive model to monitor the level of electricity consumption, detect unsual situations, and regulate the habit of consumption to save electricity.

### 6.5 Future Work

In the future, the research study could be extended in some directions to increase the accuracy of prediction and enhance the capability of implementation in the electricity economy. First of all, the expansion of data scope in many households or many geographical areas helps to assess more accurately the difference in the behavior of electricity consumption and improve the generalization of models. Besides that, applying economic or social variables such as income, the members of each family, and information about holidays or events helps the models to reflect deeper on the reasons for electricity consumption from the economic outlook. Moreover, future research could be extended by using advanced predictive models, particularly models based on the transformer structure for time series. Those model was proven to have the capability of catching the long-term relation and complex structure in data better than conventional models such as LSTM and bring higher efficiency in prediction with multiple variables. Another important improvement is implementing multi-step forecasting for a period of time, such as 6 hours, 12 hours or 24 hours, supporting the planning and regulatory electricity in reality. In addition, integrating an anomaly detection mechanism in the model helps to detect problems sooner, such as unsual consumption, damaged or leaking equipment.

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